2D Shearing Note

Definition

Shearing slants a shape by shifting one coordinate in proportion to the other.

Matrix Forms (Homogeneous 3x3)

X-shear

- Equation: x' = x + shx*y, y' = y
- Matrix:
  | 1 shx 0 |
  | 0 1 0 |
  | 0 0 1 |

Y-shear

- Equation: x' = x, y' = y + shy*x
- Matrix:
  | 1 0 0 |
  | shy 1 0 |
  | 0 0 1 |

Parameter Meaning

- shx: X-shear factor
- shy: Y-shear factor

Interpretation:

- Positive factor slants one way
- Negative factor slants the opposite way
- Large absolute factors create stronger slant

Worked Example

Apply X-shear with shx = 0.5 to P = (8, 6):

- x' = 8 + 0.5*6 = 11
- y' = 6

So P' = (11, 6)

Apply Y-shear with shy = -0.25 to same point:

- x' = 8
- y' = 6 + (-0.25)*8 = 4

So P' = (8, 4)

Python GUI

Script: tutorials/lineDrawing/09_2d_shearing.py

Controls:

- M: toggle X-shear / Y-shear mode
- Left/Right: decrease/increase shear factor
- Shift + Left/Right: fine step
- R: reset
- ESC: quit

What to Observe in GUI

- One axis value stays unchanged depending on mode
- Shape area and angles usually change under shearing
- Matrix HUD updates with shear factor every frame

Common Mistakes

- Confusing shear with rotation
- Applying X-shear formula while in Y-shear mode
- Forgetting that zero shear factor gives identity transform

Quick Practice

1. Set sh = 0 and verify transformed shape matches original.
2. Compare sh = 0.5 and sh = -0.5 in both modes.
3. Keep increasing |sh| and describe distortion pattern.